

Awesome Chocolates

Sales Analysis Dashboard - Project Report

Built with Power BI Desktop, DAX and Power Query

1. Project overview

This project is an interactive sales dashboard for a fictional chocolate company, **Awesome Chocolates**. It analyses sales, profit, costs, boxes and shipments across six countries (Australia, Canada, India, New Zealand, UK and USA) and three product categories (Bars, Bites and Other). The dashboard lets a manager see the overall picture in one glance, then drill into a country, a product category, a product or a sales person.

2. Objectives

- Track the main business numbers (sales, profit, shipments, costs, boxes) in one place.
- Show how the latest month compares with the previous month.
- Show the monthly trend and whether a month is above or below a normal month.
- Find the best and worst products and sales people by profit percentage.
- Monitor the share of small shipments (under 50 boxes) against a 10% target.

3. Data and tools

Item	Details
Dataset	ac-sample-data (Awesome Chocolates sample data provided with the course/tutorial)
Fact table	shipments (one row per shipment: sales, costs, boxes, date, product, sales person, country)
Dimension tables	calendar, products, people, locations
Helper tables	_Measure (stores all measures), Measure Selector (metric switcher)
Tools	Power BI Desktop, DAX, Power Query

4. Dashboard preview

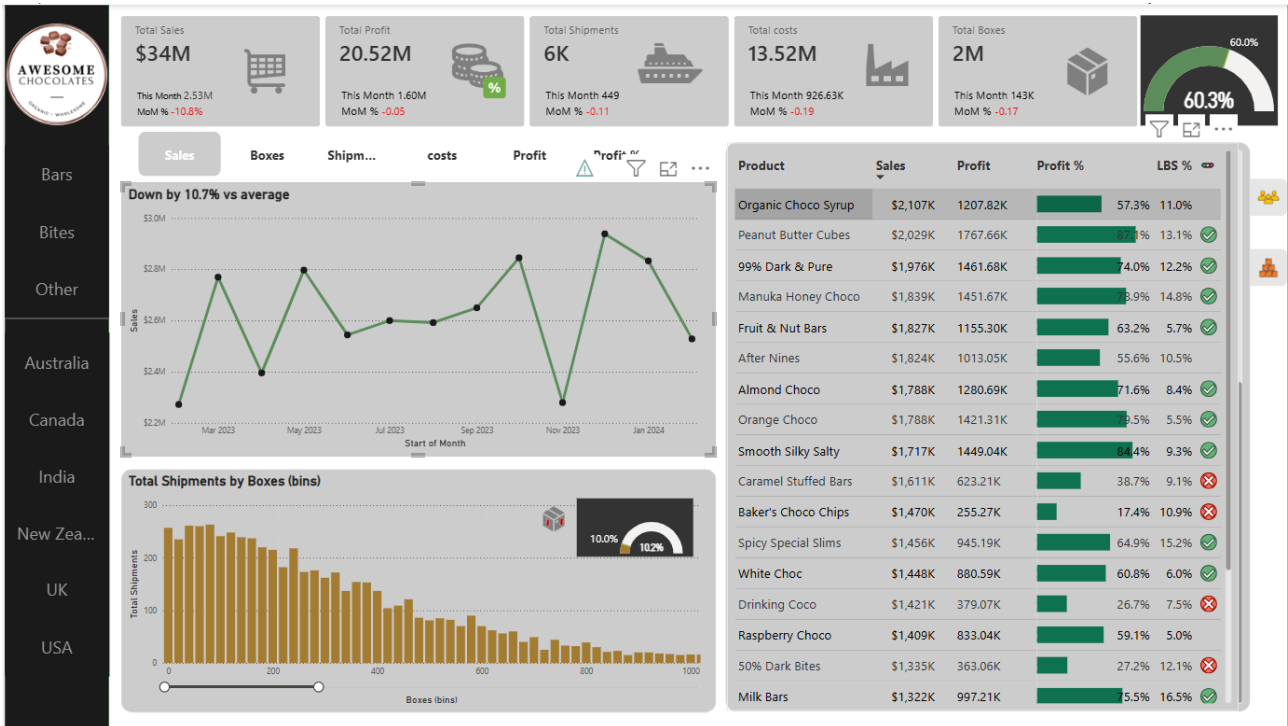


Figure 1: Main dashboard, product view (country and category buttons on the left, KPI cards on top).

5. Headline numbers

KPI	Total	Latest month	MoM change
Total Sales	\$34M	2.53M	-10.8%
Total Profit	20.52M	1.60M	Down
Total Shipments	6K	449	Down
Total costs	13.52M	926.63K	Down
Total Boxes	2M	143K	Down
Profit %	60.3% (target 60%)	-	-

All values are taken from the KPI cards of the dashboard with no country filter applied.

6. Key features

6.1 KPI cards

Five cards show the total for sales, profit, shipments, costs and boxes. Each card also shows the value for the latest month and the month-over-month (MoM) change, coloured red when the value fell. A gauge on the right shows the overall profit percentage (60.3%) against the 60% target.

6.2 Country and category navigation

The black panel on the left has buttons for the product categories (Bars, Bites, Other) and the six countries. Clicking a button filters every visual on the page, including the tooltip.

6.3 Metric switcher

The tabs above the trend chart (Sales, Boxes, Shipments, costs, Profit, Profit %) change what the line chart shows. This is built with a *Measure Selector* field parameter, so one chart serves six metrics.

6.4 Trend chart with interactive text

The line chart shows the metric by month. The text above it (for example, "Down by 10.7% vs average") is a DAX measure that compares the selected month with the average month. When a point is clicked, the text updates for that month; with nothing selected it shows the latest month.

6.5 Custom hover tooltip

Hovering over a point on the Total Boxes trend chart opens a report-page tooltip. It contains a card with the month's total boxes and a donut chart with the split of boxes by country. The tooltip respects the country buttons.

6.6 Shipment size histogram and LBS gauge

The histogram groups shipments by number of boxes (bins). The small gauge shows the share of small shipments (LBS, fewer than 50 boxes): **10.2%** against a **10.0%** target.

6.7 Product and sales person tables

The table on the right lists sales, profit, profit % (with data bars) and LBS %. Icons show whether a row meets the profit target (green tick, red cross). A button on the right switches the table between products and sales people.

7. Sales person view

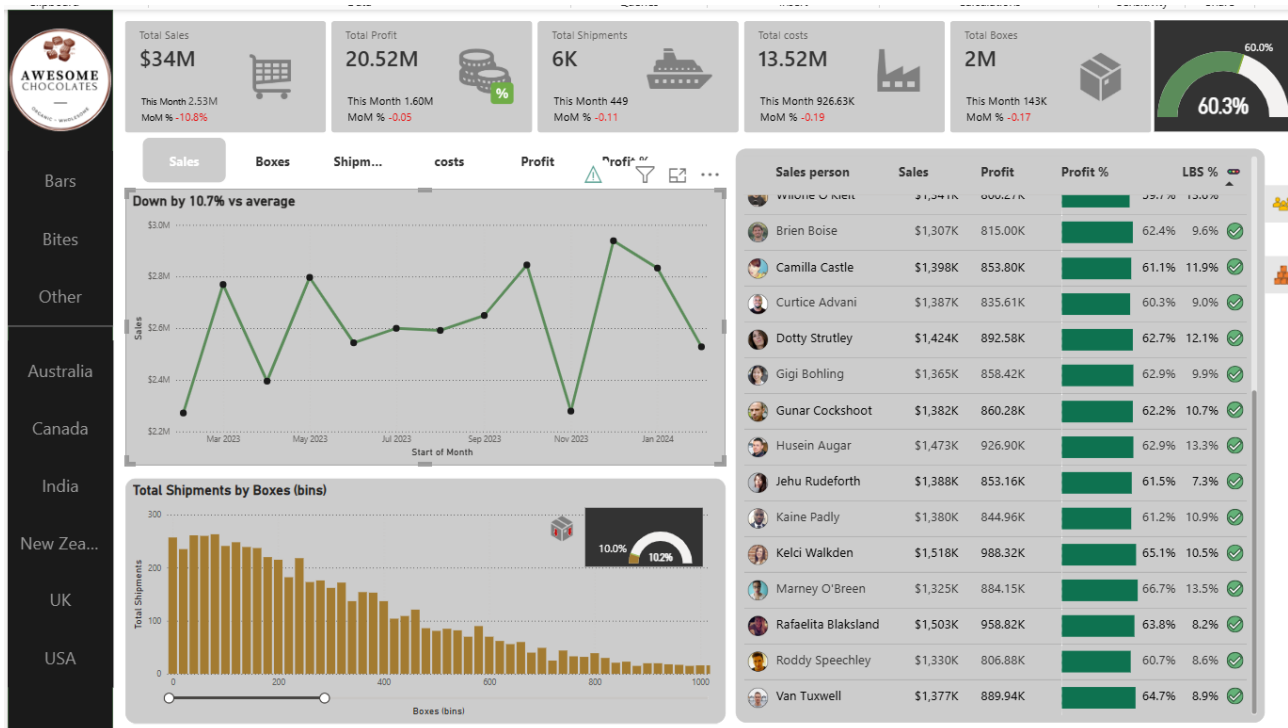


Figure 2: Same dashboard with the table switched to sales people.

8. Observations from the dashboard

- **Profit margin is on target.** Overall profit is 60.3% of sales, just above the 60% target.
- **Latest month is weaker.** The latest month shows sales of 2.53M, down 10.8% on the previous month, and the trend chart labels it 10.7% below the average month. Shipments, costs and boxes also fell.
- **Sales fluctuate month to month.** Monthly sales move between about 2.3M and 2.9M. The lowest points are around Feb 2023 and Nov 2023 (about 2.3M) and the highest is around Dec 2023 (about 2.9M).
- **Top products by sales:** Organic Choco Syrup (about \$2.1M), Peanut Butter Cubes and 99% Dark & Pure (about \$2.0M each).
- **Weak profit products:** Baker's Choco Chips (17.4%), Drinking Coco (26.7%), 50% Dark Bites (27.2%) and Caramel Stuffed Bars (38.7%) sit far below the 60% target and are flagged with red icons.
- **Sales people are close together.** Profit % ranges from roughly 54% to 67%, with sales of about \$1.3M to \$1.5M each. The lowest few are flagged red for profit.
- **Small shipments are near the limit.** LBS % is 10.2% against a 10.0% target, so it needs watching.

9. DAX and data model

All measures are stored in the `_Measure` table. They are built in layers: base measures first, then measures that reuse them. The full list with explanations is in [measures.md](#).

Group	Examples	Purpose
Base	Total Sales, Total costs, Total Boxes, Total Shipments	SUM / COUNTROWS on the shipments table
Derived	Total Profit, Profit %	Profit = sales - costs; margin = profit / sales
Latest month	Total Sales Latest Month, latest Date	Values for the newest month in the calendar
Latest MoM %	Latest MoM Sales Change %	Latest month vs the month before (KPI cards)
MoM %	MOM Boxes change %	Selected month vs previous month using PREVIOUSMONTH
LBS	LBS Count, LBS %	Share of shipments under 50 boxes
Indicator	Profit Target Indicator	2 = above target, 1 = within 10%, 0 = below
Interactive text	Boxes Change Text	"Up/Down by x% vs average" for the selected month

Example measure: Boxes Change Text

```
Boxes Change Text =
VAR _m = MAX('Calendar'[Start of Month])
VAR _cur = CALCULATE([Total Boxes], REMOVEFILTERS('Calendar'),
    'Calendar'[Start of Month] = _m)
VAR _avg = CALCULATE(AVERAGEX(VALUES('Calendar'[Start of Month]), [Total Boxes]),
    REMOVEFILTERS('Calendar'))
VAR _chg = DIVIDE(_cur - _avg, _avg)
RETURN
IF(_chg >= 0, "Up by ", "Down by ") & FORMAT(ABS(_chg), "0.0%") & " vs average"
```

How it works: it finds the selected (or latest) month, calculates that month's boxes and the average month, and turns the difference into text. REMOVEFILTERS keeps the average calculated over all months, while the country buttons still filter the numbers.

10. Possible improvements

- Format the MoM values on the profit, shipments, costs and boxes cards as percentages (they currently show decimals such as -0.05 and -0.11).
- Add a year and quarter slicer so users can compare periods.
- Add a profit-by-country page and a forecast for the next few months.
- Publish the report to the Power BI Service so it can be shared as a live link.

11. Conclusion

The dashboard turns raw shipment data into a clear, interactive view of company performance. It shows that the business meets its 60% profit target overall, that the latest month was weaker, and that a few products have profit margins far below the target. Custom touches such as the interactive change text, the metric switcher and the hover tooltip make it easier to explore than a static report.